Secure Healthcare Enterprise Portal - التقرير الأول

Submitted by:

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Report 1: Kick-off & Requirements
التقرير الأول: التأسيس وتحديد المتطلبات

Project: Secure Healthcare Enterprise Portal | **Period:** Weeks 1-2

المشروع: منصة الرعاية الصحية الآمنة للشركات | **الفترة:** الأسبوع 1-2

1. Member 1: Problem Statement & Project Goals

1.1 Problem Statement

Healthcare organizations handle large volumes of highly sensitive data such as Electronic Health Records (EHR), patient identities and staff data. They face three main threats: data breaches by outside attackers, unauthorized access by staff to records outside their duties, and transmission of data over unencrypted channels.

تتعامل المؤسسات الصحية مع كميات كبيرة من البيانات شديدة الحساسية مثل السجلات الطبية الإلكترونية وهوية المرضى وبيانات الموظفين. وتواجه ثلاثة تهديدات رئيسية: اختراق البيانات من جهات خارجية، ووصول الموظفين غير المصرح لهم إلى سجلات لا تخص عملهم، ونقل البيانات عبر قنوات غير مشفرة.

This project designs a secure environment using strong encryption, multi-factor authentication (MFA) and Role-Based Access Control (RBAC) to reduce unauthorized data exposure and cyber attacks.

يصمم هذا المشروع بيئة آمنة باستخدام تشفير قوي ومصادقة متعددة العوامل (MFA) وتحكم بالوصول حسب الأدوار (RBAC) للحد من تسرب البيانات والهجمات السيبرانية.

1.2 Project Goals

#	Goal	Measure
G1	Confidentiality	Sensitive data encrypted at rest and in transit
G2	Integrity	Digital signatures and hashing detect any modification
G3	Availability	Authorized staff can access data without interruption
G4	Strong authentication	MFA, password policy and account lockout
G5	Restricted permissions	RBAC based on least privilege

1.3 Report 1 Schedule

Week	Activity	Owner
1	Define idea, problem statement and roles	All (led by Member 1)
1	Security scope, requirements, data classification	Members 2 and 3
2	Communication channels and team structure	Member 4
2	GitHub repository and issue tracker	Member 5
2	Review, merge and submit	Member 1

2. Member 2: Security Scope and Requirements

2.1 Security Scope

In scope: the staff web portal (doctors, nurses, administration, laboratory), the medical records and staff database, the channels between browser, server and database, and the authentication, access control and audit mechanisms. Out of scope: physical building security, connected medical devices (IoT) and external billing systems.

ضمن النطاق: بوابة الويب لموظفي المستشفى (أطباء، ممرضون، إدارة، مختبر)، وقاعدة بيانات السجلات الطبية والموظفين، وقنوات النقل بين المتصفح والخادم وقاعدة البيانات، وآليات المصادقة والتحكم بالوصول والتدقيق. خارج النطاق: الأمن الفيزيائي للمباني، والأجهزة الطبية المتصلة (IoT)، وأنظمة الفوترة الخارجية.

2.2 Functional Requirements

ID	Requirement
FR1	Staff log in with an individual account
FR2	View and edit medical records according to user role
FR3	Upload lab results and link them to the patient file
FR4	Audit log for every access operation
FR5	System administrator manages users and roles

2.3 Security Requirements

ID	Requirement	Related Goal
SR1	Encrypt stored data with AES-256	Confidentiality
SR2	Encrypt transport with TLS 1.2 or higher	Confidentiality
SR3	MFA for all staff	Authentication
SR4	Store passwords hashed with salt	Confidentiality
SR5	Lock account after 5 failed attempts	Availability / protection
SR6	Role-Based Access Control (RBAC)	Confidentiality
SR7	Digital signature on critical medical reports	Integrity
SR8	Log and monitor all login attempts	Accountability

3. Member 3: Data Classification & Sensitive Assets

Data Category	Examples	Sensitivity	Required Protection
Patient medical data	Diagnosis, medication, lab results, X-ray images	Restricted	Encryption at rest and in transit, RBAC, audit
Patient identity	Name, national ID, address, phone	Confidential	Encryption, restricted access
Staff data	Salary, employee ID, contract	Confidential	HR access only
Credentials	Hashed passwords, MFA secrets	Restricted	Hashing, salt, isolation
System logs	Login logs, audit logs	Internal	Protection from modification
Public information	Working hours, department directory	Public	None

Classification principle: the higher the sensitivity level, the stricter the controls, and no user receives access beyond what their job requires.

مبدأ التصنيف: كلما ارتفع مستوى الحساسية زادت الضوابط، ولا يُمنح أي مستخدم وصولاً يتجاوز ما تتطلبه مهمته.

4. Member 4: Team Structure & Communication Channels

4.1 Team Structure

Member	Role	Main Responsibility
1	Project Manager & Scrum Master	Plan, deliverables, merging reports
2	Security Risk & Compliance Analyst	Risk analysis and CIA triad
3	Cryptography & Data Protection Engineer	Encryption and transmission protocols
4	Identity & Authentication Specialist	Authentication, passwords, MFA
5	Access Control & Systems Administrator	Git repository and RBAC

4.2 Communication Channels

Channel	Use	Frequency
Group chat	Daily coordination and quick alerts	Daily
Weekly video meeting	Progress review and task assignment	Weekly
GitHub Issues	Task and bug tracking	Continuous
Shared folder	Report drafts	Continuous

4.3 Working Rules

Each member submits their section two days before the deadline so Member 1 can review it. Every change to the repository goes through a branch and a pull request. Any blocker is reported immediately in the group chat.

يسلّم كل عضو قسمه قبل الموعد النهائي بيومين ليراجعه العضو 1. كل تعديل على المستودع يتم عبر فرع وطلب دمج (Pull Request). وأي عائق يُبلَّغ عنه فوراً في المجموعة.

5. Member 5 (Practical): Git Repository Setup & Issue Tracker

5.1 Steps

1. Create a GitHub account if you do not have one.

2. Create a repository named `secure-healthcare-portal` with a README.
3. Add the other four members as collaborators (Settings, then Collaborators).
4. Create labels `report-1` to `report-5`, plus `bug` and `documentation`.
5. Create an Issue for each task and assign it to its owner.
6. Create a Project board with three columns: To Do, In Progress, Done.
7. إن لم يكن موجوداً GitHub افتح حساباً على.
8. أنشئ مستودعاً باسم `secure-healthcare-portal` مع ملف README.
9. أضف الأعضاء الأربعة الآخرين كمتعاونين (Settings ثم Collaborators).
10. أنشئ تصنيفات `report-1` إلى `report-5` بالإضافة إلى `bug` و `documentation`.
11. أنشئ Issue لكل مهمة وعيّنها لصاحبها.
12. أنشئ لوحة Project بثلاثة أعمدة: To Do و In Progress و Done.

5.2 Local Commands

```
git clone https://github.com/<username>/secure-healthcare-portal.git
cd secure-healthcare-portal
mkdir -p reports crypto auth rbac docs
git add .
git commit -m "Initial project structure"
git push origin main
```

5.3 Suggested Issues

#	Task	Owner	Label
1	Write problem statement and goals	Member 1	report-1
2	Define security scope and requirements	Member 2	report-1
3	Classify sensitive data assets	Member 3	report-1
4	Set up communication channels	Member 4	report-1
5	Create GitHub repo and board	Member 5	report-1
6	Threat analysis and CIA mapping	Member 2	report-2
7	OpenSSL encryption demo	Member 3	report-3
8	MFA OTP implementation	Member 4	report-4
9	RBAC implementation	Member 5	report-5

5.4 Screenshots



